untiringly all my life. To this very day, instead of the definition I just mentioned, one often finds irreversibility defined as “An irreversible process is one which cannot take place in the opposite direction.” This formulation is insufficient. For it is quite possible to conceive of a process which cannot take place in the opposite direction but which can in some fashion be completely reversed.

Since the question whether a process is reversible or irreversible depends solely on the nature of the initial state and of the terminal state of the process, but not on the manner in which the process develops, in the case of an irreversible process the terminal state is in a certain sense more important than the initial state—as if, so to speak, Nature “preferred” it to the latter. I saw a measure of this “preference” in Clausius’ entropy; and I found the meaning of the Second Law of Thermodynamics in the principle that in every natural process the sum of the entropies of all bodies involved in the process increases. I worked out these ideas in my doctoral dissertation at the University of Munich, which I completed in 1879.

The effect of my dissertation on the physicists of those days was nil. None of my professors at the University had any understanding for its contents, as I learned for a fact in my conversations with them. They doubtless permitted it to pass as a doctoral dissertation only because they knew me by my other activities in the physical laboratory and in the mathematical seminar. But I found no interest, let alone approval, even among the very physicists who were closely concerned with the topic. Helmholtz probably did not even read my paper at all. Kirchhoff expressly disapproved of its contents, with the comment that the concept of entropy, whose magnitude could be measured by a reversible process only, and therefore was definable, must not be applied to irreversible processes. I did not succeed in reaching Clausius. He did not answer my letters, and I did not find him at home when I tried to see him in person in Bonn. I carried on a correspondence with Carl Neumann, of Leipzig, but it remained totally fruitless.

However, deeply impressed as I was with the importance of my self-imposed task, such experiences could not deter me from continuing my studies of entropy, which I regarded as next to energy the most important property of physical systems. Since its maximum value indicates a state of equilibrium, all the laws of physical and chemical equilibrium follow from a knowledge of entropy. I worked this out in detail during the following years, in a number of different researches. First, in investigations on the changes in physical state, presented in my probationary paper at Munich in 1880, and later in studies on gas mixtures. All my investigations yielded fruitful results. Unfortunately, however, as I was to learn only subsequently, the very same theorems had been obtained before me, in fact partly in an even more universal form, by the great American theoretical physicist Josiah Willard